How-To Library

Informatica

Performance Tuning Guidelines for PowerExchange for Microsoft Azure SQL Data Warehouse on Databricks

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Abstract

When you use PowerExchange for Microsoft Azure SQL Data Warehouse to read data from or write data to Microsoft Azure SQL Data Warehouse, multiple factors such as hardware parameters, database parameters, application server parameters, and Informatica mapping parameters impact the adapter performance. You can optimize the performance by tuning these parameters appropriately. This article describes general reference guidelines to help you tune the performance of PowerExchange for Microsoft Azure SQL Data Warehouse for Azure Databricks.

Supported Versions

- PowerExchange for Microsoft Azure SQL Data Warehouse 10.2.2

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Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance and monitor and adjust the results as required.

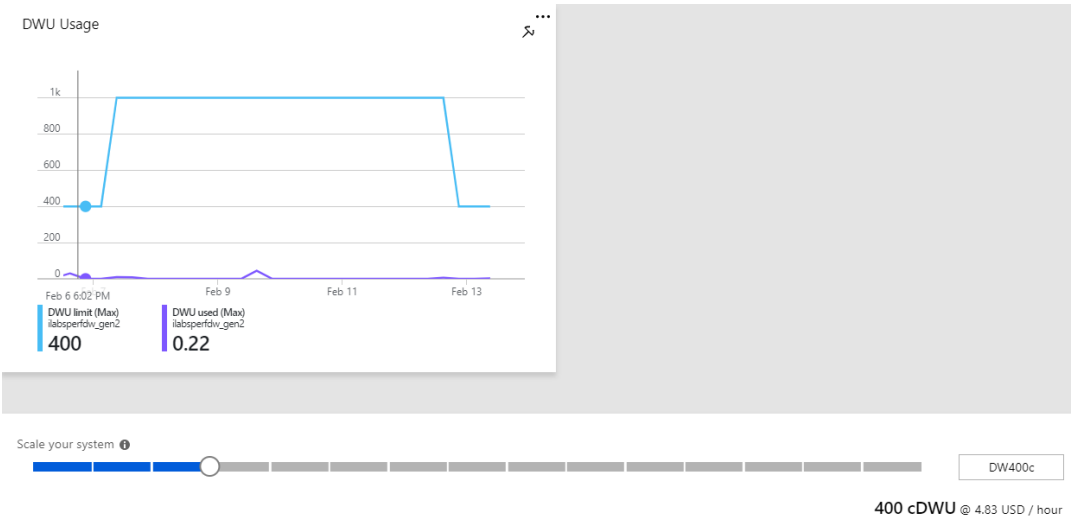
This document describes the key hardware, Databricks cluster and Informatica mapping parameters that you can tune to optimize the performance of PowerExchange for Microsoft Azure SQL Data Warehouse for Databricks .
Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of PowerExchange for Microsoft Azure SQL Data Warehouse might vary based on individual environments and other parameters even when you use the same data.

Best Practices for Tuning Performance

Consider the following best practices for better performance :

- Reading or writing AVRO files is faster than reading or writing flat files on Databricks environment.
- The Domain Host, Source, Target and Databricks cluster should be in the same region to avoid network latency.
- General-purpose V2 provides better computing and networking and is a recommended storage type by Microsoft.
- Azure SQL Data Warehouse Gen2 provides better performance than Azure SQL Data Warehouse Gen1.
- Scaling Data Warehouse Units in SQL Data Warehouse provides better performance. More the Data Warehouse Units, better the performance of Polybase read and write operations.

The following image shows the effects of scaling Data Warehouse Units:



- If the data volume exceeds 350GB, it is mandatory to scale Data Warehouse Units to 1000 to provide enough resources for Polybase to run.
- Ensure that **Enable autoscaling** is not selected while configuring the Databricks cluster.
- Maximum number of worker nodes are available to complete the job while configuring the Databricks cluster. The following image describes how to disable autoscaling and provide number of worker nodes before launching the Databricks cluster:



- Based on the resource requirement, choose the appropriate instance type for Worker Type and Driver Type.

  For more information on pricing and resource details for various instance types offered, see https://azure.microsoft.com/en-in/pricing/details/databricks/.

Tune the Mapping

You can optimize the performance of Microsoft Azure SQL Data Warehouse for Azure Databricks by tuning mapping.

You can tune the following parameters at the mapping level in the Developer Tool:

- Ports precision
- Data type mapping

For more information, see the *Informatica Performance Tuning Guide*.

Ports Precision

Precision is the maximum number of significant digits for numeric data types, or maximum number of characters for string data types. For numeric data types, precision includes scale.

You can tune the precision in Model repository mappings.

When mappings contain ports with a larger precision than required, the mapping performance degrades. Informatica recommends that you set the precision judiciously for all source ports, transformation ports, and target ports. For instance, if a string port can handle data of a maximum of 200 characters, set the precision to 200. Do not set the precision to a high value such as 1000.

Data Type Mapping

When the Integration Service reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Integration Service writes data to a target, it converts the transformation data types to the comparable native data types.

When you map source ports to transformation ports and then to target ports, avoid unnecessary data type conversions. For instance, do not map a port of the string data type to a port of the date data type. Ensure that you map ports to the same data type in all components of the mapping. Also, remove all unconnected ports from the mapping.

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